ARTHA LABS

PERSONAL RESEARCH INTELLIGENCE SYSTEM

SESSION SUMMARY & INITIAL PRODUCT BLUEPRINT

OBSERVE → ORGANIZE → INVESTIGATE → PRIORITIZE

A working record of the product vision, four source wings, initial source shortlist, observation workflow, prototype phases, and future scaling path.

DOCUMENT	STATUS	DATE
Internal session summary	Working blueprint - decisions and proposals	01 November 2026

Working product identity

Artha Labs is used here as the umbrella project identity. The application concept is described as a Personal Research Lab / Problem Solver OS.

Purpose of this document

Consolidate what has been discussed so far into an actionable internal reference. It records the agreed direction and initial shortlist; it is not a claim that integrations have already been implemented or that every source grants free production access.

Prepared for internal product planning and prototype execution.

01 / EXECUTIVE SUMMARY

The product in one view

Artha Labs is envisioned as a dynamic personal research intelligence system that acts as a continuously evolving research laboratory. It should help the user discover real-world problems, understand their context, gather evidence, connect knowledge across sources, and convert promising questions into experiments and engineering work.

Core product thesis

This is not simply a notes app, news aggregator, bookmark manager, or chatbot. It is an intelligence layer that connects information to context, evidence, decisions, prototypes, and outcomes.

Primary operating loop

STEP	WHAT HAPPENS	OUTPUT
OBSERVE	Capture signals from four source wings and direct experience.	Raw source items and observations
ORGANIZE	Normalize, tag, deduplicate, and link related records.	Structured observation inbox
INVESTIGATE	Examine provenance, corroboration, contradictions, and existing approaches.	Evidence-backed problem candidates
PRIORITIZE	Compare attention, evidence confidence, persistence, and exposure.	Ranked investigation queue
FORMULATE	Turn selected candidates into research questions and hypotheses.	Research briefs and experiment plans
LEARN	Connect implementation and evaluation outcomes back to the knowledge base.	Reusable knowledge and improved decisions

Design principles established in the discussion

- Problem-first, technology-second: choose AI, software, data, or geospatial methods only after understanding the problem.
- Capture broadly, interpret cautiously: an observation is not automatically a verified problem or a project opportunity.
- Traceability by default: preserve the original source, timestamps, attribution, and the evidence behind every conclusion.
- Human-in-the-loop: the system may classify, connect, summarize, and rank; the user retains the ability to inspect and correct it.
- Practical before autonomous: prove the basic research workflow before adding agents, complex knowledge graphs, or large-scale automation.

Scope boundary

The first prototype is a single-user research workspace with source capture, a common inbox, initial ingestion connectors, and transparent problem-candidate prioritization. Notion/GitHub integrations, autonomous research agents, large-scale crawling, and sophisticated knowledge-graph features are future extensions rather than MVP dependencies.

02 / SOURCE ARCHITECTURE

Four observation wings

The four wings are independent entry points into one shared observation system. They represent different ways in which problems become visible: people discussing lived experience, publications documenting events and research, institutions exposing how systems work, and direct personal observation.

WING	PRIMARY QUESTION	SIGNAL TYPE	CORE OUTPUT
1. Social Intelligence	What are people experiencing, discussing, struggling with, or working around?	Public discussion, complaints, user stories, community patterns	Human signals and recurring unmet needs
2. Publications & Research	What has been reported, measured, discovered, or identified as unresolved?	Journalism, newsletters, technical analysis, papers	Documented events, evidence, methods, research leads
3. Existing Systems & Institutional Intelligence	How do current systems operate, and where do they fall short?	Government portals, datasets, audits, public infrastructure, products	Operational gaps, institutional constraints, system performance
4. Self-Audit & Direct Observation	What friction or unusual behaviour is visible in everyday life?	Personal experience, conversation, workflow, field observation	Context-rich, often undocumented observations

Why all four are necessary

- Social media is a direct, high-frequency indicator of user experience and community-level needs. It can surface small, local, or niche problems before they appear in formal publications. It is a signal layer, not a representative census.
- Publications and research add context, technical depth, documented events, methods, and evidence that can be compared over time.
- Existing systems reveal formal workflows, policy intent, institutional performance, administrative bottlenecks, and datasets that social posts may not expose.
- Self-audit captures practical friction that may never be written up, indexed, or reported. It protects the system from digital-source bias.

Architecture decision

Do not build four disconnected databases. Each wing should create or reference records in one shared Observation Inbox. The same underlying problem can accumulate observations and evidence from several wings.

Classification is based on source role, not just platform

A platform can serve more than one role. For example, a Medium post may be a personal/community signal or a technical publication. A PIB release is a publication item, but it belongs to Wing 3 when used as an official institutional record. Preserve the source identity and classify the record by why it is being used.

03 / WING 1

Social Intelligence

Objective: capture direct signals of user experience, public and community dissatisfaction, emerging needs, recurring complaints, workarounds, and debates. The working shortlist is Reddit, Discord, and X, with Bluesky proposed as an optional low-friction automated addition.

SOURCE	WHY IT IS USEFUL	INITIAL ACCESS / INGESTION	MVP DECISION
Reddit	Long-form user experiences, topic-specific communities, recurring complaints, detailed comments.	Official Data API with OAuth only where the proposed use is permitted and access is approved.	Primary social source for manual discovery; automated ingestion is policy/access-gated.
Discord	Niche communities, developer groups, student groups, specialist discussions.	Bot installed in explicitly authorized servers. Message Content is a privileged intent and may require approval.	Selective integration only; no broad Discord-wide monitoring.
X	Fast-moving public debates, policy reactions, viral complaints, emerging events.	Official X API uses consumption-based / pay-per-use access. Manual URL capture first.	Keep in source registry; defer automated reads until cost and value are justified.
Bluesky (optional)	Public social data with accessible protocol services and JSON WebSocket streaming.	Public API / Jetstream stream; filter to relevant collections and topics.	Candidate first automated social connector; evaluate relevance and coverage.

Social signals to collect

- Relevant post / thread count and unique participating accounts, where available.
- Available likes, replies, reposts, shares, and other engagement indicators.
- Discussion momentum across fixed windows (for example, 24 hours, 7 days, and 30 days).
- Independent communities participating in the discussion and geographic scope where supportable.
- The original post URL, platform ID, source timestamp, retrieval timestamp, and source-specific metadata.

Important limitations and interpretation

- Social engagement is not equivalent to truth, severity, or population affected. A viral debate can be misleading, while a low-volume community problem can be genuine and persistent.
- Repeated reposts or copied claims must not be counted as independent reports. Bots, coordinated activity, platform audience differences, and controversy can distort raw engagement.
- Reddit's current developer guidance says the official route for research with Reddit data is its Reddit for Researchers programme; general developer API use for research is not automatically authorized. Treat this as a gate before automated collection.
- Discord access is server-specific and permission-based. X automation has a direct usage-cost implication. Use manual capture as the reliable fallback for both.

Wing 1 operating rule

Treat social media as a high-value user-experience sensor. Capture broadly, measure attention transparently, and corroborate claims before elevating them to established problem statements.

04 / WING 2

Publications & Research Intelligence

Objective: capture documented current affairs, investigative reporting, technology developments, industry analysis, and scholarly research. Wing 2 is divided into journalism/news, technology publications, and academic literature.

Initial shortlist

CATEGORY	INITIAL SOURCES	INGESTION APPROACH	ROLE IN THE SYSTEM
Newspapers / current affairs	The Indian Express; Business Standard	Publisher RSS feeds, configured by topic/section. Store metadata, URL, date, and permitted excerpt.	Document incidents, policy effects, regional concerns, and emerging public issues.
Technology magazines / newsletters	Digit; selected technology and industry newsletters	Digit initially via manual URL capture. Add publisher RSS or permitted newsletter feeds where available.	Track technology, product releases, engineering developments, and industry changes.
Research discovery	OpenAlex; Crossref; arXiv	Official REST / Atom APIs for metadata search and record retrieval.	Discover papers, resolve DOI metadata, and track preprints and research methods.
Manual academic discovery	Google Scholar	Human discovery; copy title, DOI, URL, and citation into Artha Labs. Do not rely on scraping Scholar pages.	Useful discovery interface; not the automated ingestion backbone.
News API option	NewsAPI	Not selected as the initial production dependency. Developer plan is for development/testing and carries delay and request limits.	Reassess only if paid access is justified and licensing fits the intended use.

Source-specific rationale

- The Indian Express has a broad RSS directory, including India, cities, governance, technology, AI, science, research, and climate sections. Its RSS terms specify personal/non-commercial use; do not republish article content without permission.
- Business Standard offers category feeds including India, industry, agriculture, economy, finance, technology, and AI. Use its official RSS directory and test feed availability before enabling schedules.
- OpenAlex is the primary broad scholarly metadata discovery source. Crossref is a supporting bibliographic/DOI metadata service. arXiv adds targeted access to computer science and other preprint literature.
- Google Scholar remains valuable for human-led discovery, but a supported, stable public ingestion API is not part of this initial design. Avoid brittle scraping.

What to extract from research records

Title, authors, publication date, DOI, venue, abstract when available, keywords/topics, publisher, URL, paper type (including preprint status when known), cited identifiers, methodology notes, findings, limitations, and relevance to a project or research question.

Copyright and source-use rule

For news and magazines, start with metadata, links, and permitted excerpts. Do not scrape paywalled pages or store/re-distribute full article text unless the source license or permission allows it.

05 / WINGS 3 AND 4

Existing Systems & Institutional Intelligence

Wing 3 is not another news aggregator. It is a structured registry of public systems, institutional processes, official datasets, audits, and documented performance. The initial geographical emphasis is India, with international comparators added only when they help explain a system or method.

SOURCE	WHAT IT CONTRIBUTES	INITIAL INGESTION	PRIORITY
data.gov.in	Central discovery point for Indian government datasets across public domains.	API where the dataset exposes one; otherwise download and register CSV / resource metadata. API key generation is available to registered users.	High
Press Information Bureau (PIB)	Official government releases, schemes, policy announcements, infrastructure and programme updates.	Official RSS feed. Treat as an institutional statement, not independent verification.	High
CPGRAMS / DARPG dashboard	Public grievance system and administrative performance indicators; useful for recurring service issues.	Public dashboard and reports/manual capture first. Do not assume access to individual grievance records or a public API.	High
CAG Digital Reports	Audit reports and findings concerning finances, programme implementation, governance, and public service delivery.	Official report pages and PDF capture; extract report metadata and relevant findings.	Medium
India Meteorological Department (IMD)	Weather forecasts, rainfall forecasts, warnings, and climate-related operational data.	Documented public API endpoints where terms and access permit. Domain-specific expansion.	Medium
CPCB / OGD AQI datasets	Air quality readings, monitoring station metadata, and environmental indicators.	OGD resource API / CSV where available. Validate anomalies; source notes that live instrument data may contain errors.	Medium

Why Wing 3 matters

- It allows comparison between what a system is intended to do, how it is formally designed, and what available performance data or audit findings suggest.
- It gives research candidates a system boundary: responsible institution, process, eligibility, service channel, data, operational constraints, and known intervention history.
- Official sources provide institutional or administrative evidence, but are not automatically neutral or complete. Pair them with independent reporting, social signals, and firsthand observations.

Wing 4 — Self-Audit & Direct Observation

Wing 4 has no external source dependency. It captures real-world friction that is practical, specific, unusual, or simply not recorded online. This is an internal, private-by-default observation channel. Design target: capture first, interpret later; a minimal form, voice-to-text option, optional photo, and offline-capable quick capture are more valuable than a long questionnaire that interrupts the moment.

CAPTURE FIELD	PURPOSE
What happened?	A short, plain-language description of the observed problem.
When / where?	Observation timestamp; optional location or setting.
Who was affected?	Stakeholder description without unnecessary personal identifiers.
What was the task?	What the person was trying to accomplish.
Where was the friction?	Delay, confusion, repeated effort, inaccessible step, failure, or unmet need.
Workaround / impact	How the person got around it; time, cost, effort, or other consequence if known.

06 / COMMON OBSERVATION WORKFLOW

One inbox, multiple sources, traceable records

Every wing should feed a common observation pipeline. Sources remain distinct, but the system uses shared record structures to search, compare, connect, and investigate observations across wings.

01	02	03	04	05
SOURCE ITEM	OBSERVATION	PROBLEM CANDIDATE	EVIDENCE REVIEW	INVESTIGATION

Four concepts must remain separate

RECORD TYPE	DEFINITION	EXAMPLE
Source item	An original post, article, paper, report, dataset, official record, or manually captured source.	A Reddit thread, CAG audit PDF, or Express article URL.
Observation	A recorded signal that something may be happening or that a need/friction may exist.	“Residents report repeated water-supply interruptions in this locality.”
Problem candidate	An inferred underlying problem grouped from one or more observations.	“Unreliable last-mile water supply and limited service transparency.”
Verified problem	A candidate whose relevant claims have been adequately substantiated for the stated purpose.	A claim supported by traceable local records, credible corroboration, and clear scope.

Initial workflow

1. Capture a source item or firsthand observation with its provenance and timestamps.
2. Normalize fields, identify duplicates, assign source wing and topic tags, and place it in the shared inbox.
3. Create a problem candidate only when an observation suggests an underlying issue; link related observations rather than copying them.
4. Attach supporting, contradicting, and contextualizing evidence. Preserve uncertainty and source independence.
5. Review candidate scope, affected stakeholders, existing systems, mitigation attempts, and unanswered questions.
6. Prioritize the candidate for further investigation; later connect research questions to experiments and results.

Timestamp discipline

Store source publication/event time, earliest known documented occurrence (if supported), time the user captured the observation, and time the application retrieved it as separate fields. The application’s first-seen date is not the problem’s birth date.

07 / SOCIAL PROBLEM PRIORITIZATION

The initial V-A-P-E model

The proposed Social Intelligence ranking model is designed to avoid equating popularity with importance. It combines attention, evidence confidence, persistence, and real-world exposure. The first version is an interpretable decision aid, not a validated measure of societal severity.

FACTOR	QUESTION	WHAT TO MEASURE	KEY SAFEGUARD
V — Visibility & Engagement	How much attention is the issue receiving, and is discussion increasing?	Relevant post volume, engagement, unique participants, independent communities, growth across fixed windows.	Deduplicate copies/reposts; compare platform-normalized signals; separate a viral spike from sustained attention.
A — Accuracy / Evidence Confidence	How well does available evidence support the claims?	Primary records, independent corroboration, credible reporting, statistical/geographic/temporal consistency, contradictions.	Do not treat missing evidence as proof of falsity. Distinguish claim confidence from problem severity.
P — Persistence & Accumulating Harm	How long and how consistently has the problem continued, and is harm accumulating?	Earliest supported occurrence, recurrence, active duration, most recent occurrence, documented trend in harm, mitigation history.	Do not infer age from the app's first-seen date; duration alone must not dominate severity.
E — Exposure	How broadly is the issue discussed, and how many people/assets are actually affected?	Separate online discussion reach from real-world population, geography, assets, service disruption, and incident exposure.	Post density is not affected-population size; demographic inference must be evidence-based and privacy-safe.

Recommended initial calculation

Provisional weighted composite

Priority Score = 0.25V + 0.25A + 0.25P + 0.25E
Each factor is normalized to 0–100. Equal weights are a transparent starting configuration only; they are not empirically validated and should remain editable.

Required score behaviour

- Show all four factor scores separately, with an explanation of how each was derived.
- Keep verification status separate: Verified, Corroborated, Unverified, or Disputed.
- Missing data is unknown, not zero. Show score completeness and either compute a clearly labelled provisional score or withhold an overall score when coverage is too weak.
- Keep online discussion reach separate from real-world exposure. Use official statistics, credible surveys, incident data, or geographic datasets for real-world exposure where available.
- Retain under-documented candidates for investigation. Low evidence confidence must not silently delete or suppress them.
- Add urgency as a separate flag/dimension for time-sensitive situations; do not conflate urgency with popularity or long-term research priority.

Three ranking views

VIEW	ANSWERS	USE
Social Attention	What is receiving attention right now?	Rank by discussion volume, engagement, and momentum.
Research / Investigation Priority	Which candidates deserve deeper investigation?	Use V-A-P-E with evidence completeness and clear score explanations.
Under-Explored Problems	What may be overlooked by popularity-driven discovery?	Surface low-visibility but persistent or significantly exposed candidates, and strong-evidence issues with limited public attention.

08 / PROTOTYPE PLAN

Three-phase implementation

The prototype is intentionally staged. Phase 1 proves the capture and organization workflow. Phase 2 adds real, permitted sources. Phase 3 adds the first transparent problem-intelligence layer.

Phase 1 — Foundation: Observation Workspace

Goal: deliver a usable local application before building complex integrations or AI agents.

COMPONENT	INITIAL SCOPE
Application shell	Overview dashboard; persistent navigation; responsive desktop/mobile UI.
Four capture workflows	Social signal capture; publication/research record; existing-system record; fast self-audit form.
Shared observation inbox	Search, filter, sort, edit, archive, link, and inspect records from every wing.
Core data model	Observation; source registry; problem candidate; evidence record; links between records.
Persistence & portability	SQLite for a single-user MVP (or retain an already suitable stack); migrations; CSV/JSON export; backup.
Operational quality	Validation; empty/loading/error states; duplicate URL warning; tests; README; .env.example; private-by-default notes.

Suggested greenfield stack discussed: React + TypeScript + Vite, Node.js + TypeScript REST API, SQLite. This is a starting option, not a mandate; inspect and preserve the existing repository stack if one exists.

Phase 1 exit criteria

- A user can create records through all four capture paths and see them persist after restart.
- Search, filters, edits, links to candidates, and CSV/JSON export work against real database records.
- No fake live metrics, non-functional buttons, or placeholder connectors are presented as working features.

Phase 2 — Source Integration: Four-Wing Ingestion

Goal: connect selected sources through a modular adapter architecture while preserving manual fallbacks.

WORKSTREAM	INITIAL SCOPE
Common adapter contract	Source configuration; fetch; parse; normalize; provenance; pagination; rate limits; retries; logs; connector health.
Low-friction feeds	RSS/Atom ingestion for The Indian Express, Business Standard, and PIB, subject to source terms.
Scholarly metadata	OpenAlex, Crossref, and arXiv search/retrieval for metadata and available abstracts.
Institutional data	data.gov.in API/CSV where exposed; CPGRAMS dashboard/report capture; CAG report registry/PDF; later IMD and CPCB datasets.
Social integrations	Reddit only through an authorized route for the intended use; Discord for approved servers/permissions; X manual first, automated only if justified; Bluesky as optional connector.
Source management	Registry, connector status, credentials, schedule, last sync, errors, manual refresh, enable/disable.

09 / PROTOTYPE PLAN CONTINUED

Phase 3 — Problem Discovery & Prioritization

Goal: convert linked observations into inspectable problem candidates and a transparent investigation queue. The user remains the final decision-maker.

CAPABILITY	INITIAL IMPLEMENTATION
Candidate workflow	Create, edit, merge, split, link, archive, and review problem candidates from observations.
Evidence management	Attach supporting, contradicting, and contextual evidence; record source independence and verification state.
Initial extraction	Use deterministic rules first; optional configured LLM for title/summary/stakeholder/topic/question proposals, with provenance and editable output.
V-A-P-E scoring	Four independent 0–100 factor scores, provisional equal weights, configurable weights, explanation, completeness, and unknown-data handling.
Three rankings	Social Attention; Research / Investigation Priority; Under-Explored Problems.
Research brief	Problem statement, stakeholders, established evidence, unknowns, existing approaches, research questions, data needs, evaluation metrics, limitations, and sources.
Evaluation set	Clearly labelled synthetic scenarios for viral/weak evidence, persistent/local, new/urgent, repeated claims, low visibility/high harm, and undocumented observations.

Core data entities

ENTITY	PURPOSE / IMPORTANT FIELDS
Source	Name, wing, category, homepage/feed/API URL, access method, terms/license note, auth requirement, status, last sync, errors.
Source item	External ID, URL, source publisher/platform, title, permitted excerpt, source timestamp, retrieval timestamp, metadata, attribution.
Observation	Original description, wing, capture timestamp, source link, topic, location, affected stakeholder, friction, workaround, review status.
Problem candidate	Working title, problem statement, scope, linked observations, hypothesis, existing approaches, questions, investigation status.
Evidence record	Claim, source, date, evidence type, supports/contradicts/contextualizes, source independence, verification status, notes.
Score snapshot	V/A/P/E values, weights, calculation date, completeness, missing fields, explanation, provisional/final label.

Recommended technical boundary

- Start with a modular monolith and relational database. A vector database, graph database, distributed queue, or microservice architecture is not an MVP requirement.
- Use a scheduler for incremental source refreshes, with backoff, timeouts, rate-limit handling, and logs. Keep API credentials server-side.
- Store references, metadata, permitted excerpts, and the user's own analysis. Avoid copying full articles or unrestricted platform content.
- Keep each connector independently switchable. The app must remain useful when any external source is unavailable or unconfigured.

Prototype completion standard

A working end-to-end path is more valuable than a large number of nominal integrations: capture a signal, retain its source, connect it to a candidate, attach evidence, explain its score, and export a research brief.

10 / FUTURE SCALING

Expansion path beyond the first prototype

FUTURE CAPABILITY	WHAT IT ADDS	WHEN TO INTRODUCE
Notion integration	Sync project documentation, literature notes, research briefs, task context, and decisions.	After core record model and workflows are stable.
GitHub authorization	Connect repositories, README files, issues, commits, experiments, and implementation context.	When candidate-to-prototype workflow is active. Use scoped access.
Knowledge graph	Link source → claim → observation → problem → question → dataset → experiment → result.	After enough connected records make relational navigation limiting.
Semantic search	Retrieve related observations, papers, and decisions by meaning rather than exact keywords.	After data quality, access control, and retrieval evaluation are in place.
Literature intelligence	Compare methods, datasets, limitations, results, and research directions across papers.	After scholarly metadata and paper records are reliable.
Proactive monitoring	Track selected sources and notify the user about relevant changes and emerging discussion.	After source relevance and deduplication can be measured.
Experiment management	Track hypothesis, code version, dataset, baseline, metrics, result, and decision.	When research briefs routinely turn into implemented experiments.
Research briefings	Weekly digest of new evidence, open questions, stalled candidates, and recommended next steps.	After relevance and recommendation quality are evaluated.
Scale infrastructure	PostgreSQL, object storage, job queues, observability, and multi-user permissions if scale requires them.	Only when volume, reliability, or collaboration needs justify them.

Non-functional guardrails

- Source governance: record terms, licensing, access method, rate limits, retention rules, and attribution for every connector.
- Privacy: personal notes remain private by default; avoid unnecessary collection of personal data; make external processing explicit.
- Evidence integrity: preserve original source URLs and record when a claim is supported, contradicted, or only contextualized.
- Data quality: distinguish unknown from zero; deduplicate copied claims; retain source and capture timestamps; monitor stale feeds.
- AI transparency: label generated summaries and classifications, link them to evidence, and allow user correction. Never present generated hypotheses as established facts.
- Operational resilience: source failures should not block manual capture; connectors should be pausable; export and backup should remain available.

Risks to actively manage

RISK	MITIGATION
Platform access changes / API costs	Manual capture fallback; modular connectors; periodic access review; usage budgets and rate limits.
Misinformation, coordinated activity, and popularity bias	Evidence states; source independence; contradiction tracking; separate attention from exposure and persistence; retain under-explored candidates.
False inference from social discussion density	Treat unique posters and community spread as discussion reach, not affected-population estimates.
Noisy feeds and premature architecture complexity	Curate sources and topics; phase connectors; add graph, vector, and agent layers only when a demonstrated limitation exists.

11 / WORKING DECISIONS

Current decisions and open items

AREA	CURRENT POSITION	STATUS
Product identity	Artha Labs — Personal Research Intelligence System; app concept: Personal Research Lab / Problem Solver OS.	Working name
Discovery architecture	Four wings feed one common Observation Inbox.	Agreed direction
Wing 1	Reddit, Discord, X; Bluesky is an optional candidate for easier automated streaming.	Shortlist; access review required
Wing 2	Indian Express, Business Standard, Digit (manual); OpenAlex, Crossref, arXiv; Google Scholar manual.	Initial shortlist
Wing 3	data.gov.in, PIB RSS, CPGRAMS dashboard, CAG reports; IMD and CPCB as domain-specific additions.	Initial shortlist
Wing 4	Private, quick, real-time self-audit; no external sources.	Agreed direction
Problem ranking	V-A-P-E: Visibility, Evidence Confidence, Persistence, Exposure. Separate online reach from real-world exposure.	Concept agreed; scoring needs calibration
Prototype staging	Phase 1 workspace; Phase 2 connectors; Phase 3 problem intelligence.	Agreed implementation sequence
Tech stack	React/TypeScript/Vite + Node/TypeScript + SQLite suggested only for a greenfield MVP.	Provisional; inspect repo first
Notion / GitHub	Future integrations for documentation and code/experiment context.	Future scope

Decisions to make during implementation

1. Confirm whether Artha Labs is the final product/organization name or a working title.
2. Choose the first social connector after reviewing current platform terms and access eligibility; Reddit API use for research is a specific approval gate.
3. Confirm the initial topic domains and keywords for RSS, papers, datasets, and social searches.
4. Decide local-only storage and backup strategy for the first release.
5. Agree on the observation schema and evidence status vocabulary before large-scale ingestion.
6. Define a small manually labelled candidate set to test ranking behaviour before trusting any score.
7. Inspect the actual code repository and preserve any working stack or components before starting Phase 1.

12 / IMPLEMENTATION HANDOFF

Suggested build order

ORDER	IMPLEMENTATION MILESTONE	EXIT CHECK
1	Foundation schema + manual capture for all wings	All four capture paths persist and export records.
2	Shared inbox + source registry	Search, filters, attribution, source metadata, and manual links work.
3	RSS + scholarly metadata connectors	Real feed and metadata ingestion; duplicate handling and sync logs.
4	Institutional source catalogue	OGD, PIB, CPGRAMS, CAG, IMD/CPCB registration and permitted ingestion paths.
5	Social connectors / manual fallback	Authorized, tested, permission-aware access; unsupported sources remain capturable.
6	Problem candidate and evidence workflow	Related observations link to candidates; contradictions and status visible.
7	V-A-P-E scoring and ranking views	Score explanation, missing-data rules, and synthetic evaluation pass.

First implementation sprint

Keep the first sprint focused on a complete manual workflow. External connectors should not block the core product from being useful.

TASK	IMPLEMENTATION CHECK
Inspect the repository	Confirm the current framework, routes, persistence layer, and reusable UI. Preserve working code before making architectural changes.
Freeze the first data model	Create Source, Source Item, Observation, Problem Candidate, and Evidence records with clear relationships and timestamps.
Build capture and inbox	Complete four capture paths, a common inbox, search/filter, record detail, and manual linking.
Prove persistence and portability	Restart the app, verify data survives, export CSV/JSON, and test backup/restore assumptions.

Phase 1 exit gate

A user can capture a social signal, publication, institutional-system observation, or firsthand experience; find it in one inbox; link it to a candidate; and export it - without requiring a single external API credential.

APPENDIX / SOURCE REGISTER

Initial source and documentation register

The links below are the official or primary reference points identified for the initial source shortlist. They support connector investigation and source configuration; they do not imply unrestricted access, permission to store full content, or a stable free production tier. Re-check current platform terms, pricing, and API permissions immediately before implementation.

Wing 1 — Social Intelligence

Reddit Developer Platform / Data Access Official source / documentation

Developer access and policy guidance, including research-use limitations. https://support.reddithelp.com/hc/en-us/articles/14945211791892-Developer-Platform-Accessing-Reddit-Data.

Discord Gateway documentation Official source / documentation

Gateway intents and message-content access requirements. https://github.com/discord/discord-api-docs/blob/main/developers/events/gateway.mdx.

X Developer Platform Official source / documentation

API access, product overview, consumption-based pricing entry point. https://developer.x.com/en/docs.

Bluesky Protocol Services / Jetstream Official source / documentation

Public protocol services and JSON WebSocket stream. https://bsky.network/docs/protocol-services/.

Wing 2 — Publications and Research

The Indian Express RSS directory Official source / documentation

Publisher-provided feeds by section, including technology, AI, cities, governance, climate, and more. https://indianexpress.com/rss/.

Business Standard RSS directory Official source / documentation

Feeds by India, industry, agriculture, economy, finance, technology, AI, and other categories. https://www.business-standard.com/rss-feeds/listing.

Digit Official source / documentation

Technology magazine and news source. https://www.digit.in/.

OpenAlex API reference Official source / documentation

Scholarly works, authors, sources, institutions, topics, and metadata API. https://help.openalex.org/api/.

Crossref REST API Official source / documentation

DOI and publisher-deposited bibliographic metadata. https://www.crossref.org/documentation/retrieve-metadata/rest-api/.

arXiv API User Manual Official source / documentation

Programmatic discovery of e-prints and research metadata. https://info.arxiv.org/help/api/index.html.

Google Scholar Official source / documentation

Manual scholarly discovery interface. https://scholar.google.com/.

NewsAPI pricing and terms Official source / documentation

Developer plan limits and permitted use. https://newsapi.org/pricing.

Wing 3 — Existing Systems and Institutional Sources

Open Government Data Platform India Official source / documentation

Government datasets and registered-user API access information. https://www.data.gov.in/help.

PIB RSS directory Official source / documentation

Official government press release and media feeds. https://www.pib.gov.in/ViewRss.aspx?lang=1&reg=1.

CPGRAMS / DARPG dashboard Official source / documentation

Public grievance portal and administrative information. https://pgportal.gov.in/darpgdashboard.

CAG Digital Reports Official source / documentation

Official digital audit reports and governance/finance findings. https://www.digitalreports.cag.gov.in/.

India Meteorological Department API reference Official source / documentation

Documented forecast and rainfall endpoints. https://api.imd.gov.in/public/api_reference.html.

CPCB real-time AQI on OGD Official source / documentation

Air quality data by monitoring stations; includes data quality caveats. https://www.data.gov.in/catalog/real-time-air-quality-index.

Reference review note

Source access, API pricing, platform policies, feed availability, rate limits, and licensing can change. This register is a planning baseline, not a substitute for checking the live terms and technical documentation during connector implementation.